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13/264,586	01/03/2012	Dennis Jones	029369.00007	9581

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ARENT FOX LLP
1717 K Street, NW
WASHINGTON, DC 20036-5342

EXAMINER

NEAGUE, IRINA

ART UNIT	PAPER NUMBER
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1627

NOTIFICATION DATE	DELIVERY MODE
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04/23/2014

ELECTRONIC

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

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Office Action SummaryApplication No.
13/264,586Applicant(s)
JONES, DENNISExaminer
IRINA NEAGUArt Unit
1627AIA (First Inventor to File)
Status
No

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 3 MONTHS FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

- 1) ☒ Responsive to communication(s) filed on 25 February 2014.
☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.
- 2a) ☐ This action is **FINAL**. 2b) ☒ This action is non-final.
- 3) ☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.
- 4) ☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 458 O.G. 213.

Disposition of Claims*

- 5) ☒ Claim(s) 1, 11-14 and 16-20 is/are pending in the application.
5a) Of the above claim(s) 14 and 16-20 is/are withdrawn from consideration.
- 6) ☐ Claim(s) _____ is/are allowed.
- 7) ☒ Claim(s) 1 and 11-13 is/are rejected.
- 8) ☐ Claim(s) _____ is/are objected to.
- 9) ☐ Claim(s) _____ are subject to restriction and/or election requirement.

* If any claims have been determined allowable, you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see

http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to PPHfeedback@uspto.gov.

Application Papers

- 10) ☐ The specification is objected to by the Examiner.
- 11) ☒ The drawing(s) filed on 25 February 2014 is/are: a) ☒ accepted or b) ☐ objected to by the Examiner.
Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).
Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).

Priority under 35 U.S.C. § 119

- 12) ☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

- a) ☐ All b) ☐ Some** c) ☐ None of the:
1. ☐ Certified copies of the priority documents have been received.
2. ☐ Certified copies of the priority documents have been received in Application No. _____.
3. ☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

** See the attached detailed Office action for a list of the certified copies not received.

Attachment(s)

- 1) ☒ Notice of References Cited (PTO-892)
- 2) ☒ Information Disclosure Statement(s) (PTO/SB/08a and/or PTO/SB/08b)
Paper No(s)/Mail Date 25 February 2014.
- 3) ☐ Interview Summary (PTO-413)
Paper No(s)/Mail Date. _____.
- 4) ☐ Other: _____.

Art Unit: 1627

The present application is being examined under the pre-AIA first to invent provisions.

DETAILED ACTION

Applicant's amendment of 25 February, 2014, in which claims 1 and 11 have been amended and claims 2 and 3 have been cancelled, is acknowledged.

Claims 1, 11-14 and 16-20 are pending in the instant application.

Claims 14 and 16-20 are withdrawn, as being drawn to a non-elected invention.

Claims 1 and 11-13 are being examined herein.

Information Disclosure Statement

The information disclosure statement (IDS) submitted on 25 February 2014, is acknowledged and considered.

Response to Arguments

In view of Applicant's submission of the IDS of 25 February 2014, the request to indicate the year/date for each non-patent document listed on page 2 of the IDS of 29 March 2013 is fulfilled.

In view of Applicant's amendment to the Specification of 25 February 2014, the objection to the Specification is herein withdrawn. Applicant has corrected a typographical error.

In view of Applicant's submission of a replacement drawing Figure 2 on 25 February 2014, the objection to the Drawings is herein withdrawn. Applicant has corrected the structure of phenyl alanine in Figure 2.

In view of Applicant's amendment of 25 February 2014, the objection to claim 3 is withdrawn. Claim 3 has been cancelled.

Art Unit: 1627

In view of Applicant's amendment of 25 February 2014, the objection to claims 14 and 16-20 is withdrawn. The status identifiers have been corrected.

In view of Applicant's amendment of 25 February 2014, the rejection of claims 1 and 2 under 35 U.S.C. 112, second paragraph, indefinite, is herein withdrawn. Claim 2 has been cancelled. Claim 1 has been amended to delete the recitation "a reward response".

In view of Applicant's amendment of 25 February 2014, the rejection of claims 2 and 3 under 35 U.S.C. 103(a) over Uccatadyamodakah and Nilakamala Kanda Gunah, in view of Van Der Glessen and Mukherjee, is herein withdrawn. Claims 2 and 3 have been cancelled.

Applicant's arguments (Remarks of 25 February 2014, pages 8-10) regarding the rejection of claims 1 and 11 under 35 U.S.C. 103(a) over Uccatadyamodakah and Nilakamala Kanda Gunah, in view of Van Der Glessen and Mukherjee, and the rejection of claim 12 and 13 in further view of Jones, Maletto and Chinery, have been carefully considered, but are not found persuasive at least for the reasons below:

Applicant's argues (page 8, second paragraph) that Uccatadyamodakah does not teach that *Mucuna pruriens* (or extracts thereof) induces a feeling of having already eaten, but rather Uccatadyamodakah teaches a combination product containing 11 different ingredients, one of which is *Mucuna pruriens*, in a "bolus like sweet preparation". Applicant further argues (page 8, last two lines, and page 9, lines 1-2) that, since the preparation contains honey and sugar, "one reading Uccatadyamodakah would quite rightfully assume that if anything in the formulation suppressed the appetite, it would be the sugar load." The examiner disagrees with Applicant's position and points out that the language of the instant claims "a composition comprising" is

Art Unit: 1627

inclusive of additional ingredients, including honey and sugar, in a composition of *Mucuna pruriens*.

With respect to Applicant's argument that it is not clear what is meant in Uccatadyamodakah by "satiating", the examiner maintains her position that the generally-accepted definition of satiating, in the context of a medicinal preparation taken on an empty stomach (as is the case in Uccatadyamodakah and in the instant invention), is a feeling of having eaten/ appetite suppression, and not a general feeling of well-being associated with sexual satisfaction, as eluded to by the Applicant (page 8, last paragraph).

In response to Applicant's argument that Uccatadyamodakah teaches a number of conditions treated with *Mucuna pruriens*, besides satiating, the examiner emphasizes that the fact that *Mucuna* can be used to treat other conditions does not diminish in any way the teaching by Uccatadyamodakah that *Mucuna* or an extract thereof is useful for appetite suppression or satiating.

In response to Applicant's argument that "Absolutely nothing in Uccatadyamodakah states or suggests that 50-500 mg of L-DOPA" the examiner directs the Applicant's attention to the reference by Van Der Glessen (US 2006/0165822), used in the rejection for the teaching that the L-Dopa content is an inherent property of the seeds of *Mucuna pruriens*. Specifically, Van Der Glessen teaches an L-Dopa content of 4.86% or 250 mg per sachet ([0107]) of *Mucuna* seed powder (each sachet containing 7.5 grams), which is within the range of instant claim 1.

Moreover, Van Der Glessen teaches that *Mucuna pruriens*, or an extract thereof, is used to treat metabolic and nutritional disorders.

Art Unit: 1627

In response to Applicant's argument that the examiner's reliance on Nilakamala Kanda Gunah "suffers from the same hindsight analysis" (Remarks, page 9, second paragraph), the examiner emphasizes that Nilakamala Kanda Gunah should be read in combination with Mukherjee, for teaching a method of stimulating in a subject a feeling of satiation with a therapeutic formulation comprising *Nelumbo Nucifera*.

Mukherjee clearly teaches that *Nelumbo nucifera* extract has anti-obesity activity, which is consistent with stimulating in a subject a feeling of satiation or appetite suppression. Furthermore, Mukherjee teaches that the nuciferine content is an inherent property of the *Nelumbo nucifera* seeds or extracts thereof. It is the examiner's view that determining the amount of *Nelumbo Nucifera* extract that contains 2 mg to 50 mg nuciferine, and any further optimization of the relative content of the two extracts, *Mucuna pruriens* and *Nelumbo Nucifera*, so that the content of L-Dopa and the content of nuciferine are as recited in instant claim 1, are well within the competency of the skilled artisan.

Thus, for all the reasons presented above, the rejection of claims 1 and 11-13 is maintained. Based on the amended claims of 25 February 2014, modified rejections under 35 U.S.C. 103(a) are made below.

Upon further consideration, it has been determined that, pursuant to the Supreme Court decision in *Association for Molecular Pathology v. Myriad Genetics, Inc.*, -- U.S. -- (June 13, 2013), the presently submitted claims of 25 February, 2014, are subject to new grounds of rejection under 35 U.S.C. 101. The rejections are detailed below.

Accordingly, this action is non-final.

Claim Rejections- 35 USC 101

35 U.S.C. 101 reads as follows:

Whoever invents or discovers any new and useful process, machine, manufacture, or composition of matter, or any new and useful improvement thereof, may obtain a patent therefor, subject to the conditions and requirements of this title.

Claims 1 and 11-13 are rejected under 35 U.S.C. 101 because the claimed invention is not directed to patent eligible subject matter. The claimed invention is directed to non-statutory subject matter because the claimed invention is directed to a naturally-occurring product, whether isolated or not, that is not patent-eligible pursuant to the Supreme Court decision in *Association for Molecular Pathology v. Myriad Genetics, Inc.*, -- U.S. -- (June 13, 2013).

An invention is not patent eligible subject matter when:

- (a) the claimed invention does not fall within at least one of the four categories of patent eligible subject matter recited in 35 U.S.C. 101 (process, machine, manufacture, or composition of matter); or
- (b) the claimed invention, although nominally falling within at least one of the four eligible categories, is directed to a judicial exception to 35 U.S.C. 101 (i.e., an abstract idea, natural phenomenon, or law of nature); or
- (c) the claimed invention would impermissibly cover every substantial practical application of, and thereby preempt all use of, an abstract idea, natural phenomenon, or law of nature; or
- (d) the claimed invention is directed to a naturally-occurring product, whether isolated or not, that is not patent-eligible pursuant to the Supreme Court decision in *Association for Molecular*

Art Unit: 1627

Pathology v. Myriad Genetics, Inc., -- U.S. -- (June 13, 2013). See MPEP 2106, MPEP 706.03(a).

Based upon an analysis with respect to the claims as a whole, claims 1 and 11-13 do not recite something significantly different than a naturally occurring product, namely an extract from *Mucuna pruriens* and an extract from *Nelumbo nucifera*. The features in the claims 1 and 11-13, namely addition of an extract from *Citrus aurantium* and an extract from *Griffonia simplicifolia*, or of an extract from *Camelia sinensis*, to a composition comprising an extract from *Mucuna pruriens* and an extract from *Nelumbo nucifera*, do not result in the claims as a whole reciting something significantly different than the natural extracts/products themselves. Furthermore, a method of inducing in a subject a feeling of having already eaten with a composition comprising an extract from *Mucuna pruriens* and an extract from *Nelumbo nucifera* represents a mere field of use of a composition, comprising two or more plant/natural product extracts, which is not markedly different from what exists in nature. Therefore, claims 1 and 11-13 are rejected as not being directed to patent eligible subject matter.

Claim Rejections - 35 USC § 103

The following is a quotation of 35 U.S.C. 103(a) which forms the basis for all obviousness rejections set forth in this Office action:

(a) A patent may not be obtained though the invention is not identically disclosed or described as set forth in section 102 of this title, if the differences between the subject matter sought to be patented and the prior art are such that the subject matter as a whole would have been obvious at the time the invention was made to a person having ordinary skill in the art to which said subject matter pertains. Patentability shall not be negated by the manner in which the invention was made.

Art Unit: 1627

The factual inquiries set forth in *Graham v. John Deere Co.*, 383 U.S. 1, 148 USPQ 459 (1966), that are applied for establishing a background for determining obviousness under 35 U.S.C. 103(a) are summarized as follows:

1. Determining the scope and contents of the prior art.
2. Ascertaining the differences between the prior art and the claims at issue.
3. Resolving the level of ordinary skill in the pertinent art.
4. Considering objective evidence present in the application indicating obviousness or nonobviousness.

Claims I and II are rejected under 35 U.S.C. 103(a) as being unpatentable over Uccatadyamodakah (In *Vangasena*; Shankar et al., Eds.; Bombay, India, 1996; p. 236; cited as Exhibit I in the IDS of 29 March 2013), Van Der Glessen et al. (US 2006/0165822, published 27 July 2006, cited in the IDS of 14 October 2011), Nilakamala Kanda Gunah (Kaiyadeva, Nilakamala Kanda Gunah. In *Kaiyadevanighantau*; Sharma et al., Eds.; Varanasi, India, 1979; p. 268-269; cited as Exhibit 3 in the IDS of 29 March 2013), and Mukherjee et al. (J. Pharm. Pharmacol. 2009, 61, 407-422, cited in PTO-892 of 25 October 2013).

Uccatadyamodakah teaches a method of stimulating in a subject a feeling of satiation with a therapeutic formulation comprising *Mucuna pruriens* (seeds), in an amount of 192 g, the seeds being dried and grounded in a mortar, with the addition of other ingredients and of water to make a soft paste, the formulation being administered as bolus like sweet pills in an amount of 48 grams (which reads on a therapeutically effective amount). Uccatadyamodakah teaches the administration is oral, in the evening, on empty stomach (before the subject eats a meal, as in claims I, II).

Uccatadyamodakah does not teach a method of stimulating a feeling of having already eaten with a combination of an extract from *Mucuna pruriens* and *Nelumbo nucifera*.

Art Unit: 1627

Uccatadyamodakah does not specifically teach that an extract of *Mucuna pruriens* contains L-Dopa.

Uccatadyamodakah does not specifically teach administration of the composition 45 minutes before the subject eats a meal, as in instant claim 11.

Van Der Glessen (US 2006/0165822) teaches that *Mucuna pruriens*, or an extract thereof, contains L-Dopa (Abstract, [0016], [0022]); thus, the L-Dopa content is an inherent property of the seeds of *Mucuna pruriens*. Specifically, Van Der Glessen teaches an L-Dopa content of 4.86% or 250 mg per sachet ([0107]) of *Mucuna* seed powder (each sachet containing 7.5 grams), which is within the range of instant claim 1.

Moreover, Van Der Glessen teaches that *Mucuna pruriens*, or an extract thereof, is used to treat metabolic and nutritional disorders ([0058]).

Nilakamala Kanda Gunah teaches a method of stimulating in a subject a feeling of satiation with a therapeutic formulation comprising *Nelumbo Nucifera*.

Nilakamala Kanda Gunah does not specifically teach that an extract of *Nelumbo nucifera* contains an aporphine alkaloid such as nuciferine, nor does it teach that the amount of nuciferine in the formulation is 2 mg to 50 mg.

Mukherjee teaches that *Nelumbo nucifera* extract has anti-obesity activity (page 407, under Introduction, line 17; also page 412, under Traditional uses, line 9; also page 416, under Anti-Obesity activity). Mukherjee teaches that seeds of *Nelumbo nucifera* contain alkaloids such

Art Unit: 1627

as nuciferine (page 409, left column, lines 7 and 9, and page 410, Figure 1, bottom left structure).

Thus, Mukherjee teaches that the nuciferine content is an inherent property of the *Nelumbo nucifera* seeds or extracts thereof.

It would have been obvious to a person of ordinary skill in the art at the time the invention was made to combine an extract of *Mucuna pruriens* and an extract of *Nelumbo nucifera*, in a therapeutic formulation, wherein the amount of L-Dopa is 50 mg to 500 mg and the amount of nuciferine alkaloid is 2 mg to 50 mg, and use the formulation to stimulate a feeling of having already eaten.

The person of ordinary skill in the art at the time the invention was made would have been motivated to combine an extract of *Mucuna pruriens* and an extract of *Nelumbo nucifera*, because *Mucuna pruriens* (seeds or extract) was known to induce satiation, as taught by Uccatadyamodakah, and was known to be used (in doses/ sachets of seeds containing 250 mg L-Dopa) to treat metabolic and nutritional disorders, as taught by Van Der Glessen; and

Nelumbo nucifera was known to induce a feeling of satiation in a subject, as taught by Nilakamala Kanda Gunah, and was known to contain nuciferine alkaloids and have anti-obesity activity, as taught by Mukherjee. Therefore, one of ordinary skill in the art would have reasonably expected that combining extracts from *Mucuna pruriens* and *Nelumbo nucifera*, known to be useful for the same purpose, and optimizing the ratio between the active ingredients, will result in an appetite-suppressing composition. Such an optimization of the ratio between the content of *Mucuna pruriens* and *Nelumbo nucifera* in the formulation (which corresponds to the claimed content of L-Dopa versus nuciferine alkaloid), with the aim of maximizing the

Art Unit: 1627

Uccatadyamodakah, Van Der Glessen, Nilakamala Kanda Gunah, and Mukherjee are as above.

Jones (US 6,224,873) teaches a method of weight control/loss by regulation of appetite using compositions comprising *Citrus aurantium* (Examples 2, 3, 5, columns 17-18, subjects completed a daily appetite and satiety rating questionnaire). Jones teaches (column 16, last paragraph) that the Citrus material, either in the form of an extract or as the natural material, may be given in combination with other herbs that possess beneficial effects for humans, particularly in respect to weight loss.

Maletto (US 2006/0078627) teaches (page 2, [0025], [0027]) a method of reducing hunger with a composition comprising *Griffonia Simplicifolia*. Maletto teaches that *Griffonia Simplicifolia* is a natural source of 5-hydroxytryptophan, which is a precursor to serotonin ([0027]). Maletto teaches that serotonin is one of the most important brain neurotransmitters involved in appetite suppression; it has been established that increased serotonin levels decrease appetite.

Chinery (US 2004/0077556) teaches a method for promoting weight loss in mammals with a nutritional supplement composition comprising *Camellia sinensis* ([0006]). Chinery teaches that green tea extract, *Camellia sinensis*, contains epigallocatechin gallate (EGCG), which suppresses appetite ([0006]).

It would have been obvious to a person of ordinary skill in the art at the time the invention was made to add an extract from *Citrus aurantium* and an extract from *Griffonia simplicifolia* to a combination of extracts of *Mucuna pruriens* and *Nelumbo nucifera*, in a therapeutic formulation used to stimulate a feeling of having already eaten.

Art Unit: 1627

The person of ordinary skill in the art at the time the invention was made would have been motivated to to add an extract from *Citrus aurantium* and an extract from *Griffonia simplicifolia* to a combination of extracts of *Mucuna pruriens* and *Nelumbo nucifera* and use the combination in a method of stimulating in a subject a feeling of satiation, because extracts of *Citrus aurantium* were known to suppress appetite and induce weight loss, as taught by Jones, and extracts of *Griffonia Simplicifolia* were known to reduce hunger, as taught by Maletto. The person of ordinary skill in the art would have been further motivated to add such extracts together because Jones teaches that *Citrus aurantium*, either in the form of an extract or as the natural material, may be given in combination with other herbs that possess beneficial effects for humans, particularly in respect to weight loss.

Similarly, it would have been obvious to a person of ordinary skill in the art at the time the invention was made to add an extract from *Camellia sinensis* to a combination of extracts of *Mucuna pruriens* and *Nelumbo nucifera*, in a therapeutic formulation used to stimulate a feeling of having already eaten. The person of ordinary skill in the art at the time the invention was made would have been motivated to add an extract from *Camellia sinensis* to a combination of extracts of *Mucuna pruriens* and *Nelumbo nucifera* and use the combination in a method of stimulating in a subject a feeling of satiation, because extracts of *Camellia sinensis* were known to suppress appetite and promote weight loss, as taught by Chinery.

Therefore, one of ordinary skill in the art would have reasonably expected that combining extracts from *Citrus aurantium*, *Griffonia simplicifolia*, *Mucuna pruriens* and *Nelumbo nucifera*, or combining extracts from *Camellia sinensis*, *Mucuna pruriens* and *Nelumbo nucifera*, known to be useful for the same purpose, will result in an appetite-suppressing composition. Since all

Art Unit: 1627

therapeutic effect, is well within the competency of a person of ordinary skill in the art. Since all composition components herein are known to be useful as inducing satiation, it is considered *prima facie* obvious to combine them into a single composition useful for the same purpose. At least additive effects would have been reasonably expected. See *In re Kerkhoven*, 205 USPQ 1069 (CCPA 1980).

Furthermore, it would have been obvious to a person of ordinary skill in the art at the time the invention was made to administer the combination of extracts of *Mucuna pruriens* and *Nelumbo nucifera* 45 minutes before the subject eats a meal. The person of ordinary skill in the art would have been motivated to administer the composition before the subject eats a meal, because compositions comprising *Mucuna pruriens* were known to be administered orally on an empty stomach, as taught by Uccatadyamodakah. Varying the time before the composition is administered to the subject with the aim of improving the therapeutic effect of the composition is considered routine for the skilled artisan.

As such, claims 1 and 11 are rejected as *prima facie* obvious.

Claims 12 and 13 are rejected under 35 U.S.C. 103(a) as being unpatentable over Uccatadyamodakah, Van Der Glessen, Nilakamala Kanda Gunah, and Mukherjee, as applied to claims 1 and 11 above, in further view of Jones (US 6,224,873 of 1 May 2001, cited in PTO-892 of 25 October 2013), Maletto et al. (US 2006/0078627, published 13 April 2006, cited in PTO-892 of 25 October 2013) and Chinery (US 2004/0077556, published 22 April 2004, cited in the IDS of 14 October 2011).

Art Unit: 1627

composition components herein are known to be useful as inducing satiation, it is considered *prima facie* obvious to combine them into a single composition useful for the same purpose. At least additive effects would have been reasonably expected. See *In re Kerkhoven*, 205 USPQ 1069 (CCPA 1980).

As such, claims 12 and 13 are rejected as *prima facie* obvious.

Claims 1 and 11-13 are rejected under 35 U.S.C. 103(a) as being unpatentable over Ram Hart et al. (US 2006/0030625, published 9 February 2006, cited in PTO-892) and Mukherjee et al. (J. Pharm. Pharmacol. 2009, 61, 407-422, cited in PTO-892 of 25 October 2013), in view of Jones (US 6,224,873 of 1 May 2001, cited in PTO-892 of 25 October 2013), Maletto et al. (US 2006/0078627, published 13 April 2006, cited in PTO-892 of 25 October 2013) and Chinery (US 2004/0077556, published 22 April 2004, cited in the IDS of 14 October 2011).

Ram Hart et al. (US 2006/0030625) teaches a method of elevating serotonin and dopamine levels in a subject by administering to a subject a composition comprising 390 mg of L-DOPA as velvet bean, *Mucuna pruriens* seed extract, and 5-hydroxytryptophan (5-HTP) as *Griffonia simplicifolia* seed extract ([0081], Example 1, pages 6-7).

Ram Hart also teaches a method of elevating dopamine levels by administering to a subject a composition comprising 175 mg L-DOPA as velvet bean, *Mucuna pruriens* seed extract ([0082], Example 2, page 7). Ram Hart teaches that dopamine is necessary, for example, for hunger regulation ([0017]). Ram Hart teaches administration of said composition before eating (no food was permitted until completion of the study [0084]).

Art Unit: 1627

Ram Hart does not teach a method of stimulating a feeling of having already eaten with a combination of an extract from *Mucuna pruriens* and *Nelumbo nucifera*.

Mukherjee teaches that *Nelumbo nucifera* extract has anti-obesity activity (page 407, under Introduction, line 17; also page 412, under Traditional uses, line 9; also page 416, under Anti-Obesity activity). Mukherjee teaches that seeds of *Nelumbo nucifera* contain alkaloids such as nuciferine (page 409, left column, lines 7 and 9, and page 410, Figure 1, bottom left structure). Thus, Mukherjee teaches that the nuciferine content is an inherent property of the *Nelumbo nucifera* seeds or extracts thereof.

Jones (US 6,224,873) teaches a method of weight control/loss by regulation of appetite using compositions comprising *Citrus aurantium* (Examples 2, 3, 5, columns 17-18, subjects completed a daily appetite and satiety rating questionnaire). Jones teaches (column 16, last paragraph) that the Citrus material, either in the form of an extract or as the natural material, may be given in combination with other herbs that possess beneficial effects for humans, particularly in respect to weight loss.

Maletto (US 2006/0078627) teaches (page 2, [0025], [0027]) a method of reducing hunger with a composition comprising *Griffonia simplicifolia*. Maletto teaches that *Griffonia simplicifolia* is a natural source of 5-hydroxytryptophan, which is a precursor to serotonin ([0027]). Maletto teaches that serotonin is one of the most important brain neurotransmitters involved in appetite suppression; it has been established that increased serotonin levels decrease appetite.

Chinery (US 2004/0077556) teaches a method for promoting weight loss in mammals with a nutritional supplement composition comprising *Camellia sinensis* ([0006]). Chinery

Art Unit: 1627

teaches that green tea extract, *Camellia sinensis*, contains epigallocatechin gallate (EGCG), which suppresses appetite ([0006]).

It would have been obvious to a person of ordinary skill in the art at the time the invention was made add an extract of *Nelumbo nucifera* to a composition taught by Ram Hart, and use the formulation to stimulate a feeling of having already eaten.

The person of ordinary skill in the art at the time the invention was made would have been motivated to add an extract of *Nelumbo nucifera* to a composition taught by Ram Hart, because *Nelumbo nucifera* was known to have anti-obesity activity, as taught by Mukherjee, and a composition comprising 390 mg of L-DOPA as *Mucuna pruriens* seed extract, and 5-hydroxytryptophan (5-HTP) as *Griffonia simplicifolia* seed extract was taught by Ram Hart to elevate serotonin and dopamine levels in a subject. Furthermore, since dopamine levels were known to be necessary in hunger regulation, as taught by Ram Hart, and increased serotonin levels were known to be correlated with decreased appetite, as taught by Maletto, a composition of *Mucuna pruriens* and *Griffonia simplicifolia* taught by Ram Hart was expected to suppress appetite and thus have anti-obesity activity.

Therefore, one of ordinary skill in the art would have reasonably expected that combining extracts from *Mucuna pruriens*, *Griffonia simplicifolia*, and *Nelumbo nucifera*, known to be useful for the same purpose, and optimizing the ratio between the active ingredients, will result in an appetite-suppressing composition. Such an optimization of the ratio between the content of *Mucuna pruriens* and *Nelumbo nucifera* in the formulation (which corresponds to the claimed content of L-Dopa versus nuciferine alkaloid), with the aim of maximizing the therapeutic effect, is well within the competency of a person of ordinary skill in the art. Since all composition

Art Unit: 1627

components herein are known to be useful as inducing satiation, it is considered *prima facie* obvious to combine them into a single composition useful for the same purpose. At least additive effects would have been reasonably expected. See *In re Kerkhoven*, 205 USPQ 1069 (CCPA 1980).

Furthermore, it would have been obvious to a person of ordinary skill in the art at the time the invention was made to administer the combination of extracts of *Mucuna pruriens* and *Nelumbo nucifera* 45 minutes before the subject eats a meal. The person of ordinary skill in the art would have been motivated to administer the composition before the subject eats a meal, because compositions comprising *Mucuna pruriens* were known to be administered orally on an empty stomach, as taught by Ram Hart. Varying the time before the composition is administered to the subject with the aim of improving the therapeutic effect of the composition is considered routine for the skilled artisan.

It would have been obvious to a person of ordinary skill in the art at the time the invention was made to add an extract from *Citrus aurantium* to a combination of extracts of *Mucuna pruriens*, *Nelumbo nucifera* and *Griffonia simplicifolia*, in a therapeutic formulation used to stimulate a feeling of having already eaten. The person of ordinary skill in the art at the time the invention was made would have been motivated to to add an extract from *Citrus aurantium* to the combination and use the combination in a method of stimulating in a subject a feeling of satiation, because extracts of *Citrus aurantium* were known to suppress appetite and induce weight loss, either alone or in combination with other herbs that promote weight loss, as taught by Jones.

Art Unit: 1627

Similarly, it would have been obvious to a person of ordinary skill in the art at the time the invention was made to add an extract from *Camellia sinensis* to a combination of extracts of *Mucuna pruriens* and *Nelumbo nucifera*, in a therapeutic formulation used to stimulate a feeling of having already eaten. The person of ordinary skill in the art at the time the invention was made would have been motivated to add an extract from *Camellia sinensis* to the combination, because extracts of *Camellia sinensis* were known to suppress appetite and promote weight loss, as taught by Chinery.

Therefore, one of ordinary skill in the art would have reasonably expected that combining extracts from *Citrus aurantium*, *Griffonia simplicifolia*, *Mucuna pruriens* and *Nelumbo nucifera*, or combining extracts from *Camellia sinensis*, *Mucuna pruriens* and *Nelumbo nucifera*, known to be useful for the same purpose, will result in an appetite-suppressing composition. Since all composition components herein are known to be useful in suppressing appetite, it is considered *prima facie* obvious to combine them into a single composition useful for the same purpose. At least additive effects would have been reasonably expected. See *In re Kerkhoven*, 205 USPQ 1069 (CCPA 1980).

As such, claims 1 and 11-13 are rejected as *prima facie* obvious.

Conclusion

Claims 1 and 11-13 are rejected.

Any inquiry concerning this communication or earlier communications from the examiner should be directed to IRINA NEAGU whose telephone number is (571)270-5908. The examiner can normally be reached on Monday-Thursday 7:30-5:00.

Art Unit: 1627

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, SREENIVASAN PADMANABHAN can be reached on (571)272-0629. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

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/IRINA NEAGU/

Examiner, Art Unit 1627

/SREENI PADMANABHAN/

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